

MARIE ROSE MWISENEZA

rmwisene@andrew.cmu.edu | mwisemarierose@gmail.com

+250780850683

www.linkedin.com/in/mwiseneza | <https://github.com/mwisemarierose>

PROFESSIONAL SUMMARY

Innovative tech mind capable of adapting variety of technology and software solutions and managing databases. Valuable team member has experience diagnosing problems and developing solutions.

EDUCATION

- Master's Degree in Information Technology

August 2025 - May 2027

Carnegie Mellon University Africa, Kigali

- Bachelor's Degree in Computer Engineering

January 2019 - December 2024

UNIVERSITY OF RWANDA, Kigali

EXPERIENCE

Research Assistant – Kinyarwanda ASR for Health and Agriculture Advisory

May 2026 - Present

Carnegie Mellon University, Kigali/Rwanda

- Supporting the development of Kinyarwanda ASR systems for health applications through speech data preparation, model fine-tuning, performance evaluation, and research on low-resource speech technologies, with a focus on improving recognition accuracy for domain-specific vocabulary.

Frontend Developer

January 2025 - Present

Semanticservices Rwanda - Hybrid, Kigali

- contributes to compliance automation and cloud-based enterprise solutions. Worked on frontend and backend development using React, TypeScript, Node.js, and Azure services, with experience in OSCAL/NIST compliance workflows, CI/CD automation, XML validation, API integration, and secure cloud infrastructure.

Software Developer

June 2024 - July 2025

MyDevConnect, Remote

- Developed new game features and logic to expand core gameplay.
- Built and integrated RESTful APIs for player stats, leaderboards, and matchmaking.
- Enhanced server performance and reduced response times through code optimization.

Software Developer Intern

February 2024 - June 2024

TechnoServe, Kigali/Rwanda

- Developed a mobile application using Kotlin for Android.
- Designed features to manage and track daily checklists at coffee washing stations.

PROJECTS

Keylogger Creation & Detection

October 2025 – December 2025

- Designed a consent-based cross-platform Python keylogger capturing keystrokes, timestamps, active window titles, and process telemetry while simulating attacker techniques including silent execution and controlled data exfiltration.

A Cross-Cultural Analysis of

January 2026 – May 2026

Extremism Labels in African Contexts

- Designed and conducted a perspectivist annotation study with 34 African annotators across two language conditions (English and Kinyarwanda) to evaluate cultural bias in Western extremism detection systems

BabyNest

October 2023 – December 2024

- Developed an IoT- and AI-enabled smart baby bed that monitors a child's sleep and movement patterns in real time to enhance safety and comfort.

SKILLS

Programming languages: JavaScript, Java, C, C++, Python, Kotlin | **AI/ML:** Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib |

Team collaboration tools: Git, Pivotal tracker, Asana, clickUp | **Databases:** PostgreSQL, MongoDB, MySQL, SQLite.

LANGUAGES

- English | Kinyarwanda.

CERTIFICATIONS

Certificate of CCNA from CISCO networking academy | Certificate in software development from Solvit-Africa | Certificate in software development from Andela Rwanda | Certificate of AI and IoT engineering from JICA |
Certificate of spectrum management from ITU | Certificate in foundation of Project Management from Coursera